

Arsh Hashmi

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EDUCATION

Binghamton University, State University of New York, Thomas J. Watson College of Engineering and Applied Science

B.S. Mechanical Engineering (5+1 B.S./M.S. Intended), B.S. Computer Science, Economics Minor - 3.4 GPA

Organizations: Binghamton Motorsports, Watson Combat Robotics League, American Society of Mechanical Engineers

Expected May 2029

TECHNICAL SKILLS

Engineering Software: Autodesk Fusion 360, SolidWorks, ANSYS Fluent (CFD), MATLAB, Simio

Fabrication: GD&T/Tolerancing, CAN-BUS Decoding & Embedded Sensor Systems, Carbon Fiber Composite Layups, MIG/Stick Welding, CNC/Waterjet Manufacturing, Electrical Diagnostics & Wiring Harness Repair, Rapid Prototyping, FDM 3D-Printing

Programming: Python, C/C++ (Arduino/Embedded)

Languages: English (Native), Urdu (Native), Hindi/Punjabi (Intermediate), Spanish (Working-Professional)

PROJECTS

Project ARIA - Self Driving F1 Car

East Meadow, New York

Founder & Project Lead

May 2026 - Current (In Progress)

- Founded and led a 15-person cross-university engineering team (Cornell, Vaughn, Binghamton, Stony Brook) spanning mechanical, electrical, and aerospace disciplines to design/build a fully autonomous F1 car from the ground up, repurposing salvaged components (ATV, shopping cart steel) into a custom chassis
- Led drivetrain and frame fabrication: diagnosed a failed stator and faulty CDI via multimeter testing, replaced the wiring harness, then reverse-engineered chassis geometry via 3D scanning and Fusion 360 CAD before stick-welding (E6013 rod) a custom extended steel chassis to CAD-verified tolerances
- Engineered the full electrical architecture, including CAN-BUS integration via Arduino Nano and MCP2515 modules to drive a repurposed instrument cluster, alongside relay-controlled lighting systems, GPS, and hall-effect sensor integration for real-time vehicle data
- Achieved a fully operational, drivable vehicle through iterative testing and troubleshooting across drivetrain, electrical, and structural systems; currently leading integration of autonomous driving capability as the project's next development phase

Active Variable Incidence Wing System (AVIWS)

East Meadow, New York

Solo Engineer & Designer

June 2026

- Designed, engineered, and road-tested an IMU-driven active rear wing for a BMW e93, dynamically actuating wing incidence in real time across drag reduction, low downforce, and airbrake modes activated based on live acceleration data
- Validated aerodynamic performance via CFD analysis in Fusion 360 (k-epsilon turbulence model, 30 mph) prior to fabrication, then confirmed real-world actuation behavior through on-road testing and video-documented validation
- Developed embedded C++ firmware with sensor fusion filtering (10-sample moving average) and debounce logic to ensure stable, low-latency actuation under real driving conditions
- Engineered the full electronics stack: Arduino Nano, MPU-6050 IMU (I2C), dual servo actuation, buck-converter voltage regulation, integrating sensing, control, and mechanical actuation into a closed-loop system

PROFESSIONAL EXPERIENCE

Binghamton University MakerLab

Binghamton University, New York

Additive Manufacturing/Systems Technician

February 2026 - Current

- Operated and maintained a 10-printer FDM fleet, resolving 1,000+ service requests via root-cause troubleshooting to improve first-pass yield
- Optimized print parameters (layer height, infill, temp profiles) through data-driven process analysis, minimizing material waste and reprint rate
- Provided technical mentorship on CAD modeling, 3D scanning, and soldering, supporting design feasibility and dimensional accuracy for student projects
- Advised on design-for-manufacturability (DFM) and print/scan feasibility, recommending process adjustments to improve output quality

JC Broderick & Associates

Hauppauge, New York

Project Monitor & Air Sampling Technician

May 2026 - September 2026

- NYS-certified Project Monitor & Air Sampling Technician on active asbestos abatement sites, ensuring full regulatory compliance under NYS Part 56
- Set up and calibrated air sampling pumps across sites, calculating flow rates to maintain sample volumes within regulatory limits
- Maintained chain-of-custody documentation with strict accuracy, verifying worker asbestos-handling license compliance across contractor crews
- Conducted daily site inspections and coordinated logs/communication between supervisors and contractors

Cosmic Pcs

East Meadow, New York

Founder

June 2022 - February 2025

- Built & operated an e-commerce platform generating ~\$2,000-\$3,000 monthly revenue, optimizing product selection & pricing based on demand patterns
- Designed structured cost-performance models to evaluate tradeoffs between CPU/GPU configurations under budget constraints
- Developed/utilized Excel/Gantt tracking systems for managing a 3-person team, inventory, pricing, and order flow, improving efficiency & organization
- Translated customer requirements into tailored system configurations, applying constraint-based decision making (cost/performance) to generate parts-lists

ENGINEERING EXPERIENCE

Binghamton Motorsports (SAE)

Binghamton University, New York

Engineer - Formula EV Aerodynamics Team

September 2025 - Current

- Designed nosecone & front wing assemblies in Fusion, iterating on geometry to optimize aerodynamic efficiency; performed CFD simulations in ANSYS Fluent evaluating drag/lift coefficients, velocity contours, and pressure distributions across multiple flow conditions
- Applied GD&T principles to DXF files for waterjet-cut components, ensuring dimensional accuracy & proper fitment within the assembly
- Manufactured aerodynamic components using 3K/12K carbon fiber composite layups; including mold preparation, vacuum bagging, and post-cure finishing

Watson Combat Robotics League (WCRL)

Binghamton University, New York

Engineer - Full Combat Robotics Team (Antweight Class - 11lb)

September 2025 - Current

- Designed & iterated on a 1 lb antweight combat robot in Fusion 360, applying GD&T & geometry refinements to improve structural integrity
- Soldered full electronics stack: drive motors, weapon system, and LiPo battery, to competition-legal standards; refined based on match performance data